

IN4MATX 133: User Interface Software

Lecture 19: Prototyping in Research

Announcements

- Course evaluation is up, or will be up soon
- Prof. Epstein take these seriously, as does the Department of Informatics
- Every year, we make changes to 133 in response to what students put in the evaluations

Today's goals

By the end of today, you should be able to...

- Articulate requirements needed for developing early-stage prototypes
- Consider how development requirements often shape design needs, for better or worse
- Describe research outcomes related to personal informatics

What does it take to make a prototype?

What do I mean by making a prototype?

- Moving *beyond* a mockup or interactive prototype (e.g., what you might create in Figma or Adobe XD)
 - Almost always involves some coding
 - Often requires thinking about storage (databases), user profiles, telemetry
 - We'll discuss a bit more why to make a prototype
- You might see different phrases associated with this concept
 - High-fidelity prototype, Minimum viable prototype

Question

How have you made prototypes in the past?

- ☐ A I've never sketched a design or made an app
- ☐ B I've made sketches in the past, but never made apps
- ☐ C I've made some apps before, never followed a sketch
- ☐ D I've made apps that have followed sketches
- ☐ E I've made apps with sketches, as well as databases, logins, and telemetry

How have you made prototypes in the past?

I've never sketched a design or made an app

0%

I've made sketches in the past, but never made apps

0%

I've made some apps before, never followed a sketch

0%

I've made apps that have followed sketches

0%

I've made apps with sketches, as well as databases, logins, and telemetry

0%

Four stories of design and deployment

Four stories of design and deployment

- Talk2Care
- Rebite
- RECOVER
- Privacy Sandbox

Talk2Care

Ziqi Yang, Xuhai Xu, Bingsheng Yao, Ethan Rogers, Shao Zhang, Stephen Intille, Nawar Shara, Guodong Gordon Gao, and Dakuo Wang. 2024. Talk2Care: An LLM-based Voice Assistant for Communication between Healthcare Providers and Older Adults. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 8, 2, Article 73 (June 2024), 35 pages. <https://doi.org/10.1145/3659625>

Have you ever used an LLM voice assistant?

Do you use it for your health?

Talk2Care

Context

- Older adults often have changing health conditions, but it is challenging for them to communicate with healthcare provider



- Providers have the expertise, but are often overloaded
- Existing communication rely on traditional phone calls or patient portals
- Opportunity: LLMs are good at conversations and processing information in languages

What we did

Our Process: Formative Study

- Understanding communication challenges through interviews older adults and healthcare providers



10 Older Adults



9 Healthcare Providers



Semi-structured Interview

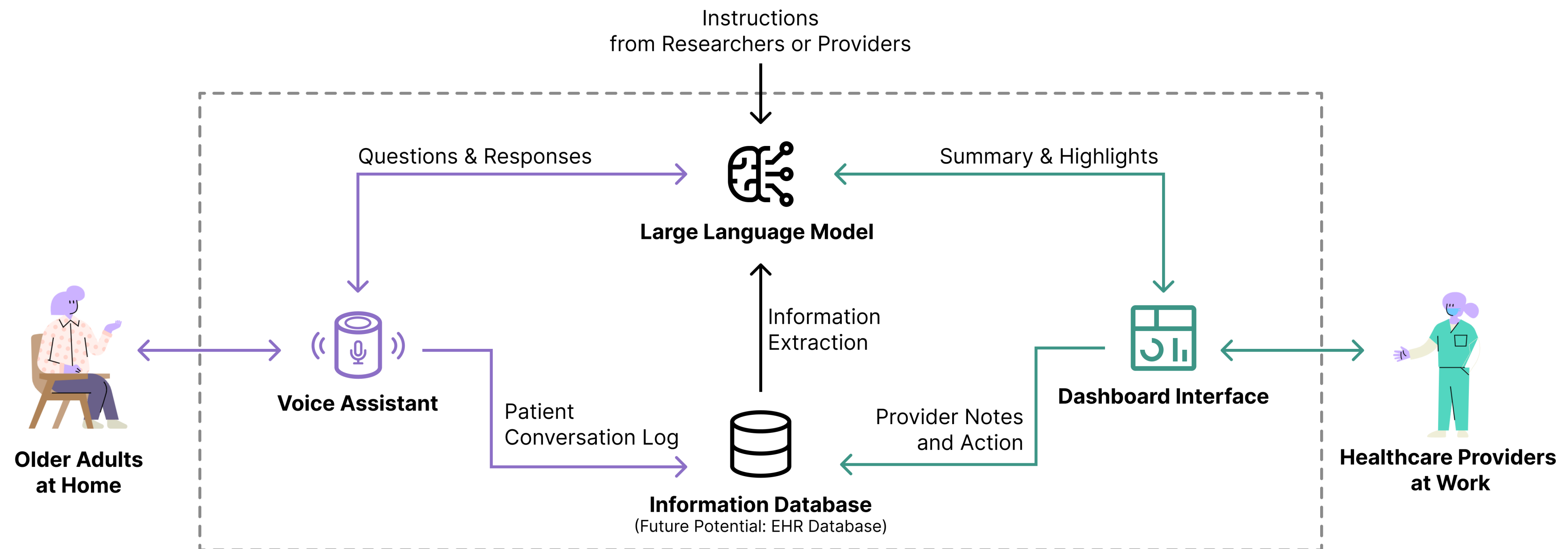


25-30 min

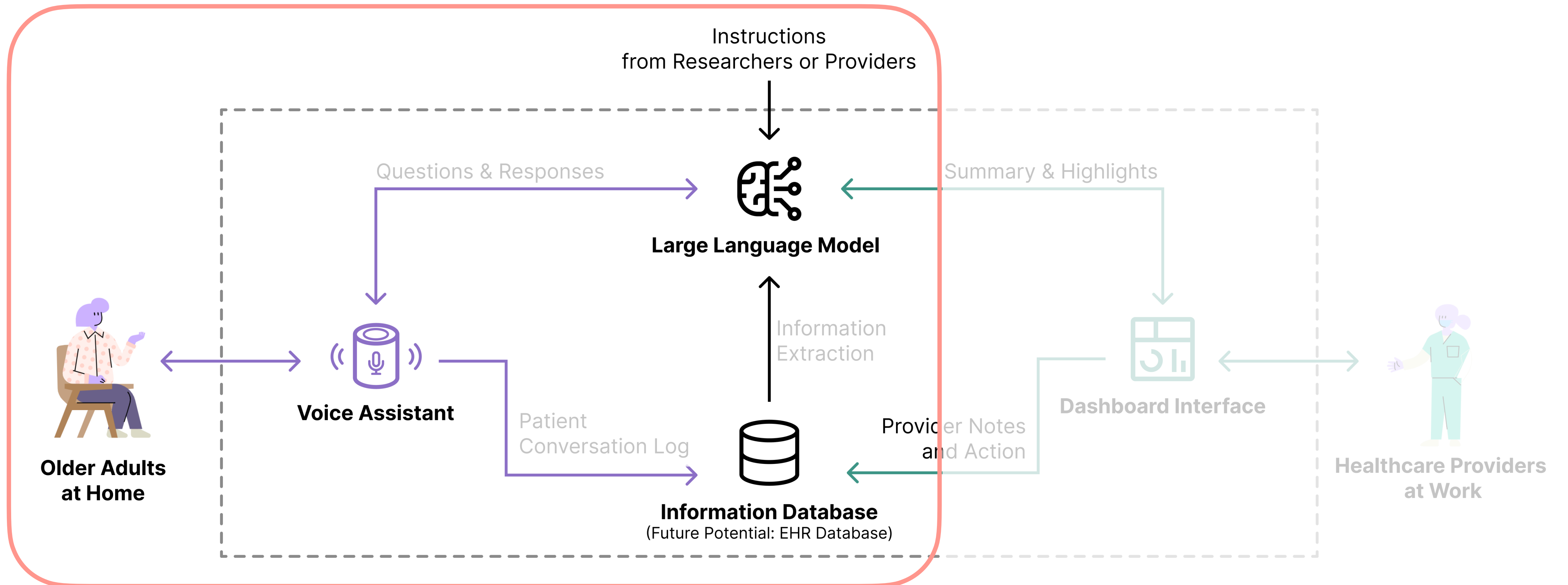
What we did

Our Process: Design and Development

- Designing and Developing Talk2Care system



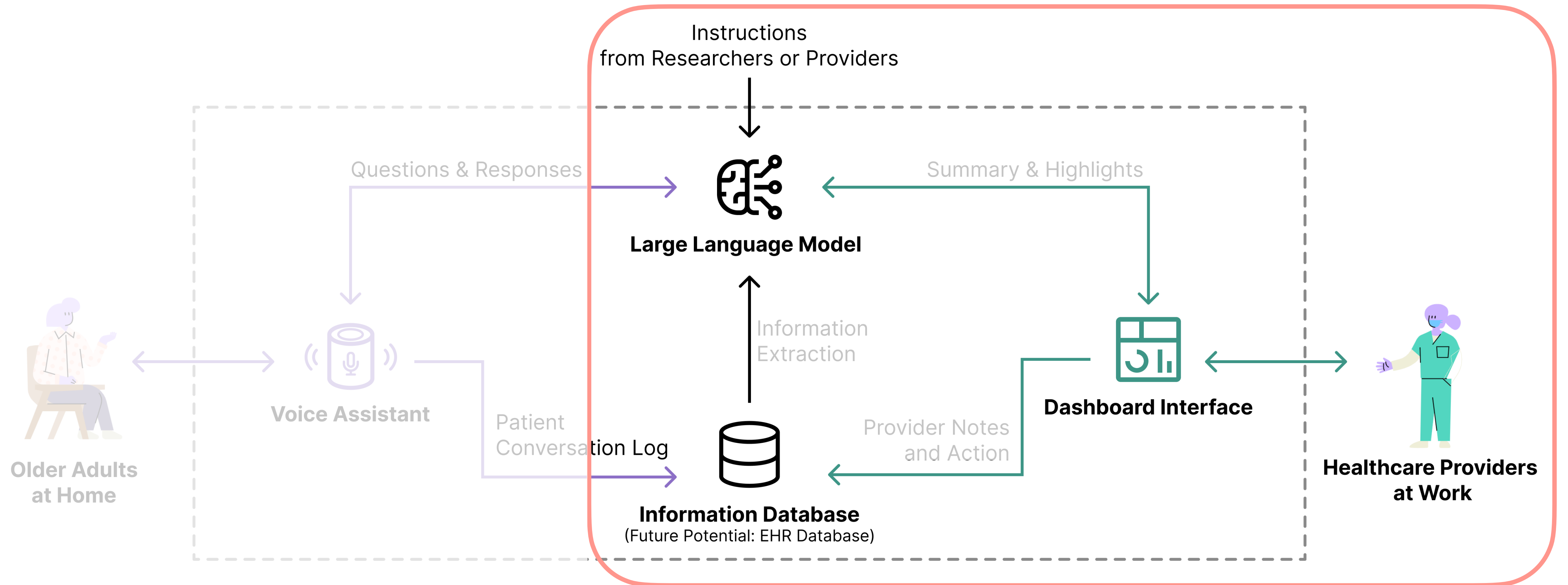
LLM VA for Older Adults





[LLM-Powered VA]

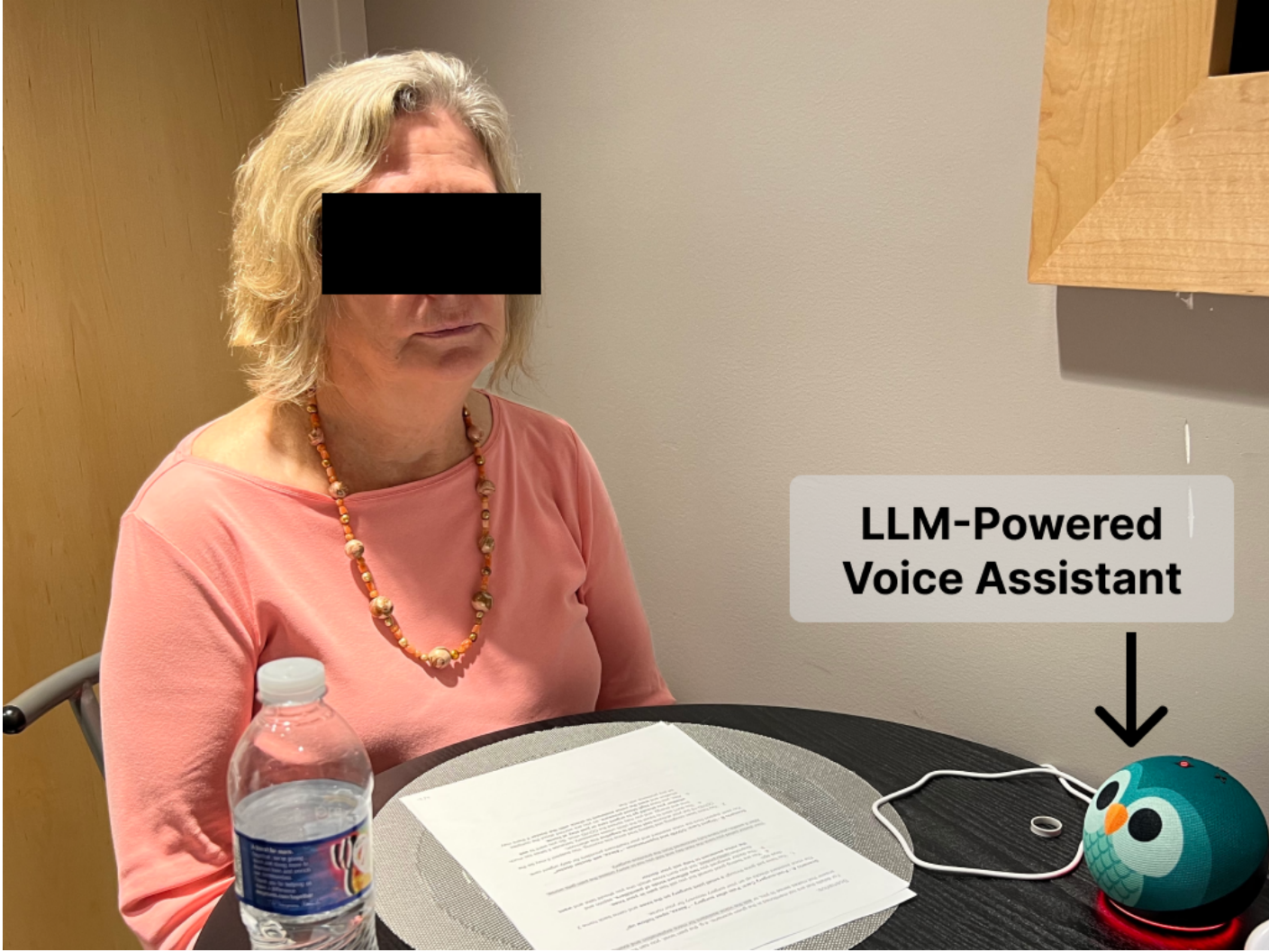
LLM Dashboard for Healthcare Providers



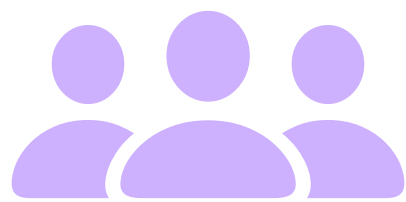


What we did

Our Process: User Studies



LLM-Powered Voice Assistant



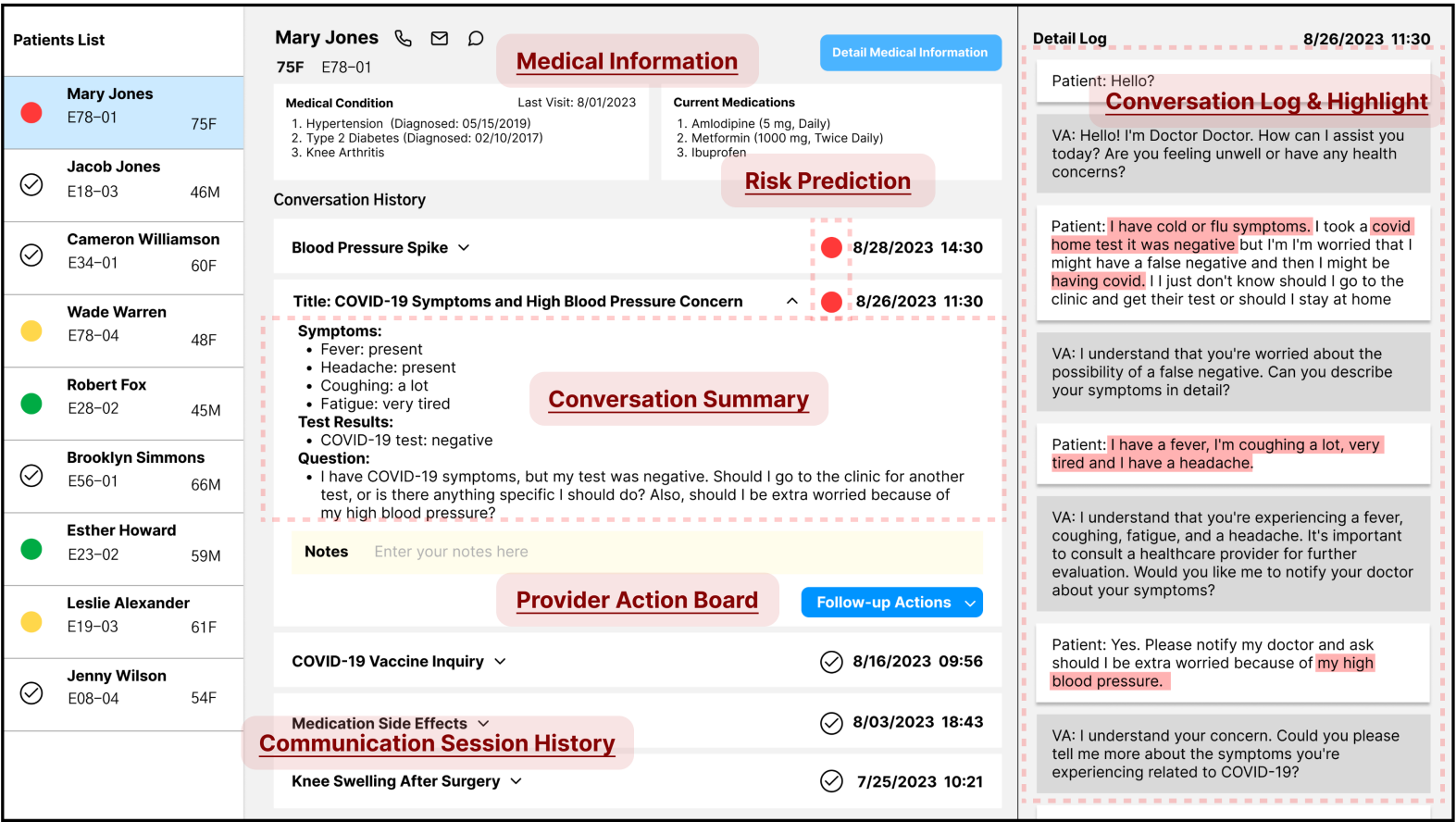
10 Older Adults



25-30 min

User Study

- Role-play in two scenarios
- Traditional method baseline
- SUS+TLX+effectiveness evaluation
- Follow-up interview



LLM-Powered Dashboard Interface



9 Healthcare Providers

Talk2Care

Specifications

- Max use: 1 user for voice assistant at a time, no limits for dashboard users
- Vue for website
- Python scripts in Alexa settings for LLM assistant
- GPT-3.5 -> GPT-4 during testing
- Built by 2 student / experienced developers over 1~2 months

Talk2Care

Why build it?

- Is it feasible to have LLM VA support older adults' daily healthcare communication needs?
- The focus was more on observing and reflecting than evaluating
 - The feasibility and experience
 - What was helpful, what was risky

Talk2Care

What was easy?

- Drafting questions and scripts for older adult conversations
- Test conversations between LLM (ChatGPT) and myself on a local computer
- Drawing a Figma prototype of a provider dashboard

Talk2Care

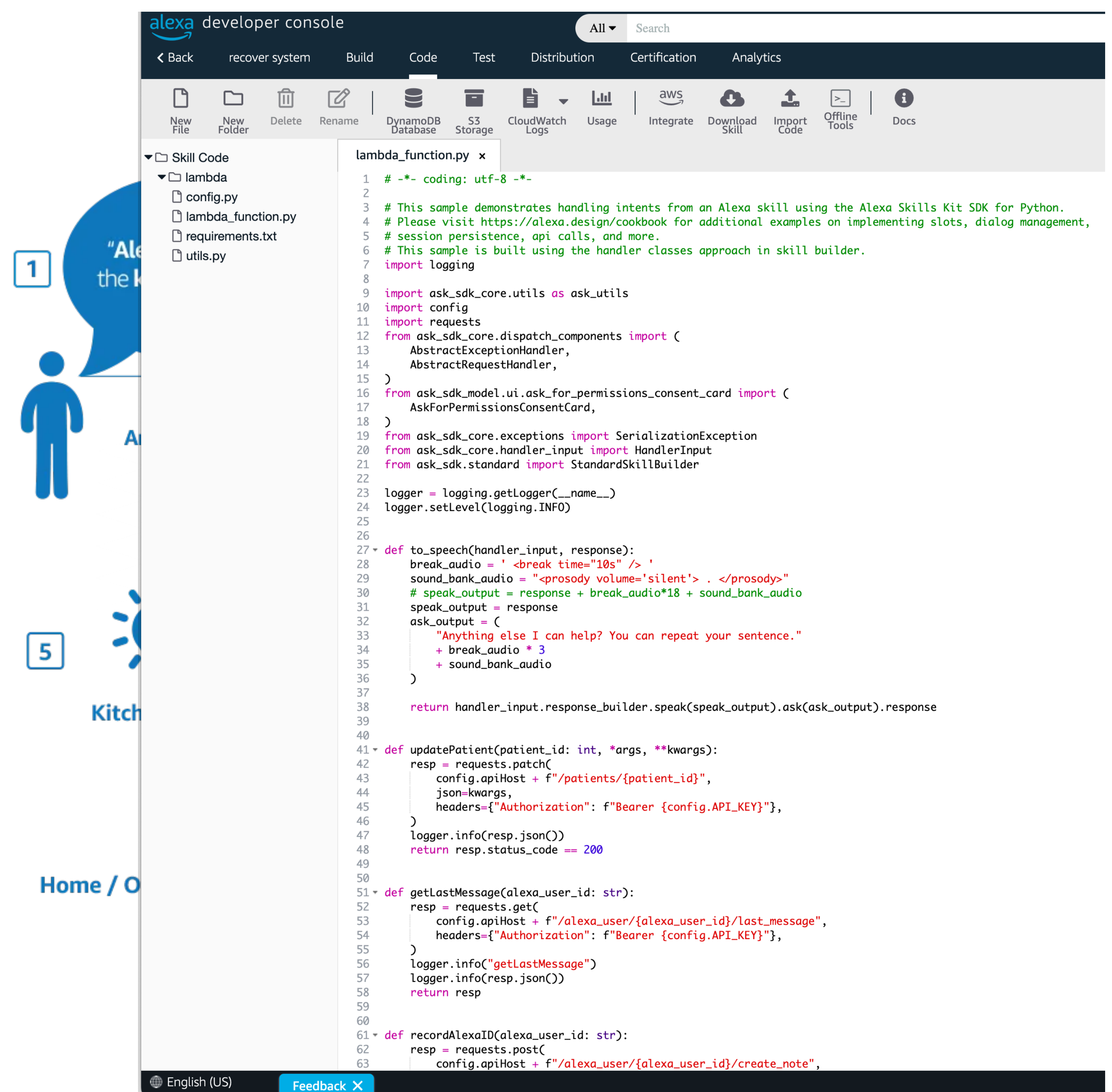
What was hard?

- Engineering efforts to connect LLM to Echo Dot (see next slide)
- Recruit older adults and set up an experiment setting (a “home”!) for them to test it
- Reflect on future directions, risks and ethical concerns using AI for healthcare and for older adults

Talk2Care

VA via Alexa

- Implementing an Alexa skill in Alexa Console
- Multi-turn conversation
- Connecting to LLM
- Alexa account configurations



Talk2Care

Major Takeaways

- We sometimes build prototypes to test out an idea
- Making things work for real users (not just the developer!) can take quite some engineering efforts
- We don't have to implement the most complete software

Rebite

Rebite

Context

Have you ever used LLMs to help with your diet goals?



1. It provides energy (good for a meal)

This meal likely includes:

- Burger (protein + fat + carbs)
- Fries (carbohydrates + fat)
- Soda (sugar/carbohydrates)

So it supplies calories and quick energy, which can help if you:

- haven't eaten much earlier
- need fuel for the afternoon
- just finished physical activity

4. A slightly better way to eat it

If you want a more balanced lunch, you could:

- keep the burger (good protein)
- eat half the fries
- replace soda with water or unsweetened drink

That keeps the meal closer to 500–700 kcal and improves nutrition.

Rebite

Context

Food journaling **goals** change over time

Weight loss → muscle gain → mindful eating

Current apps are rigid: fixed **metrics**, **structured** input, **one goal at a time**

← Add Daily Goal ✓

👑 Premium Feature

Create Your Own
Select day(s), then adjust calories or macros.

M T W T F S S

Calories
Auto-calculated by gram values below 2,060

Carbohydrates 50% 258 g

Protein 20% 103 g

Fat 30% 69 g

In reality, the way we perceive and experience food is far less structured.

Rebite

Context

Can LLMs help reframe old food data under new goals?

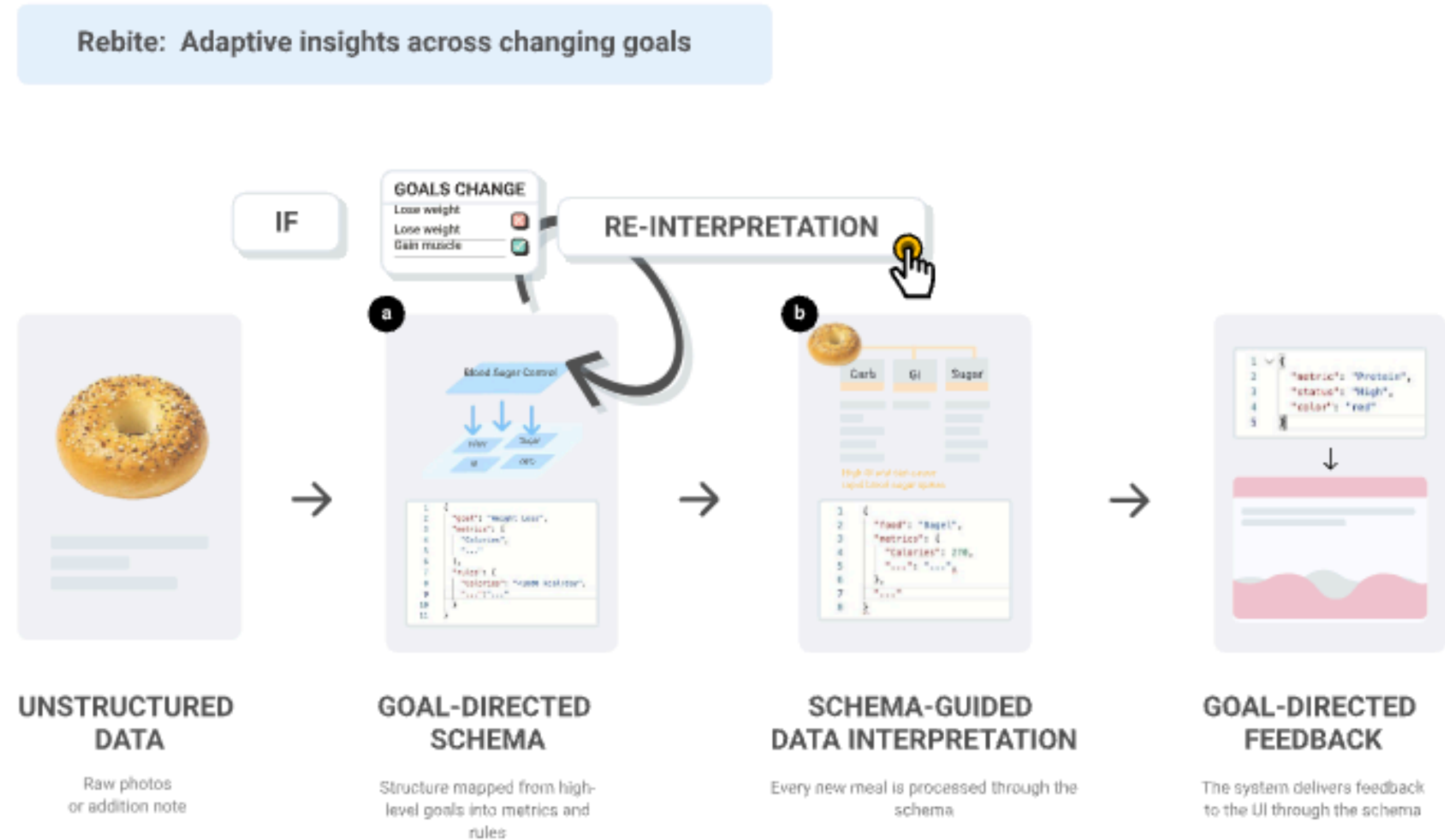
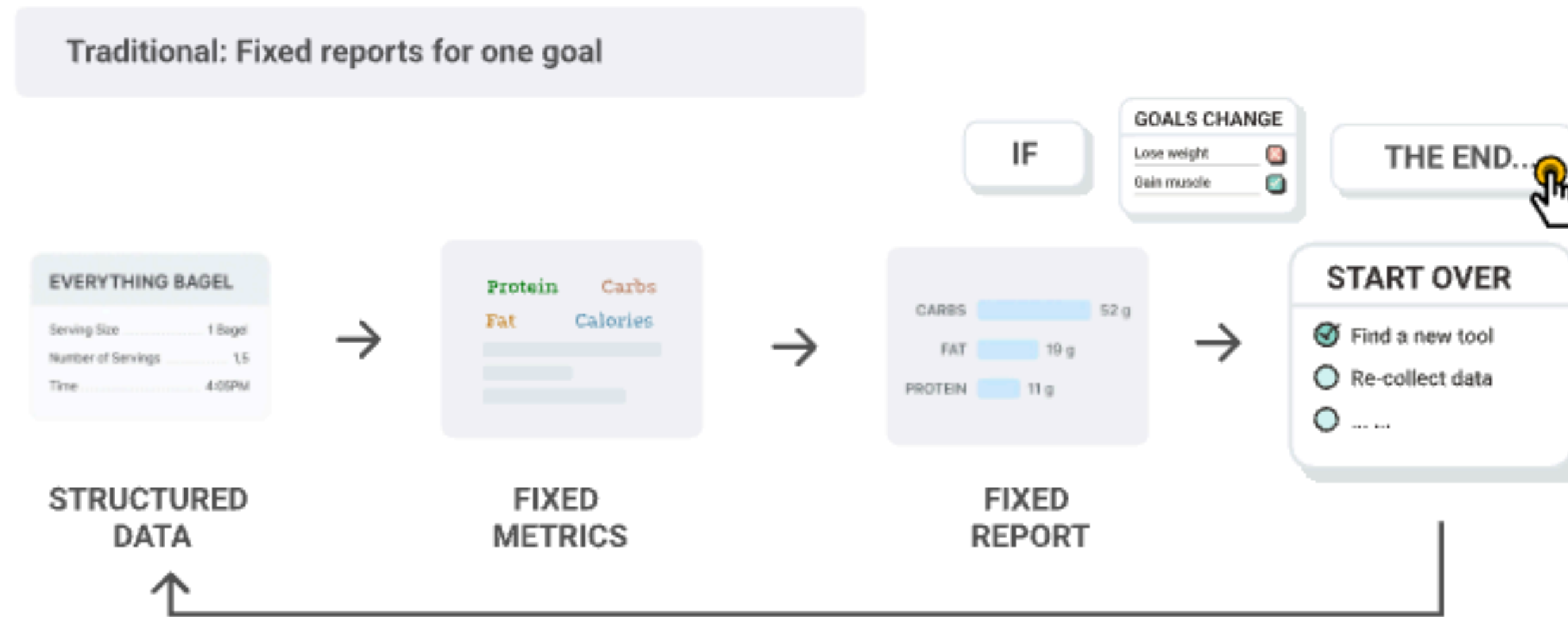
Can LLMs help us record food experiences in unstructured ways while supporting structured reflection?

Rebite

Why build it?

- The research question was about a **DYNAMIC** experience over time
 - Does reframing old data feel useful or disorienting? Can't mockup a week of use
- LLM re-interpretation is non-deterministic — can't be faked
 - A Wizard-of-Oz approach would need a nutritionist on call 24/7 per participant
- If the app feels broken, participants stop logging — and you have no data

Rebite



Rebite

How it works

Users select **high-level goals** (e.g., "lose weight", "improve dietary diversity")

System maps **goals** → concrete **metrics** using LLM (calories, protein, fiber...)

Users upload meal **photos**; LLM interprets **against** goal-metric schema

Feedback chain: **Goal** → **Metric** → **Interpretation** → **Evidence**

Rebite

What was easy

- Forking **Clean Slate (open-source tracker)** = instant foundation
 - Prototyping strategy: don't build what already exists, extend it
- LLMs collapsed months of feature work into prompt engineering
 - Photo → nutrition → feedback: **one API call replaces a whole nutrition database**

Rebite

What was hard

- **LLM output must be structured, but LLMs are stochastic**
 - Same photo, same prompt → different JSON structure. Breaks your UI.
- **Prototype quality directly affected study validity**
 - participants withdrew / Reviewer question
 - **food recognition errors eroded trust**
- **You can't tell participants "it's just a prototype, ignore the bugs"**

Rebite

LLM limitations

- **Portion estimation**
 - Could not reliably estimate portion sizes from photos
- **Hidden ingredients**
 - Photos don't reveal cooking methods, oils, or seasonings
- **Misidentification**
 - Similar-looking foods confused (e.g., rice vs. quinoa)

Rebite

Implications & design insights

- **Data should "live many lives"**
 - Personal data shouldn't be locked to one set of goals
- **Goals as outcomes, not preconditions**
 - Dynamic outcomes of reflection, continuously negotiated
- **LLM as bridge, not oracle**
 - Powerful transformation, but users must know when to correct

RECOVER

Ziqi Yang, Yuxuan Lu, Jennifer Bagdasarian, Vedant Das Swain, Ritu Agarwal, Collin Campbell, Waddah Al-Refaie, Jehan El-Bayoumi, Guodong Gao, Dakuo Wang, Bingsheng Yao, and Nawar Shara. 2026. RECOVER: Designing a Large Language Model-based Remote Patient Monitoring System for Postoperative Gastrointestinal Cancer Care. Proc. ACM Hum.-Comput. Interact. 10, 2, Article CSCW032 (April 2026), 57 pages. <https://doi.org/10.1145/3788068>
Currently available on: <https://arxiv.org/abs/2502.05740>

RECOVER

What we did

Legend



Google Docs



Conversation Log Examples



Flow Diagram



Website Figma Prototype



Question List



Website Implementation

Participants

1, 2, 3, 4

1, 2, 5

1, 2, 5

1, 2, 4, 5

1, 2, 5

1, 2, 4, 5

1, 2, 3, 4

Artifacts



Conversation Flow



Interactive Dashboard

Information Architecture

Session 1

Session 2

Session 3

Session 4

Session 5

Session 6

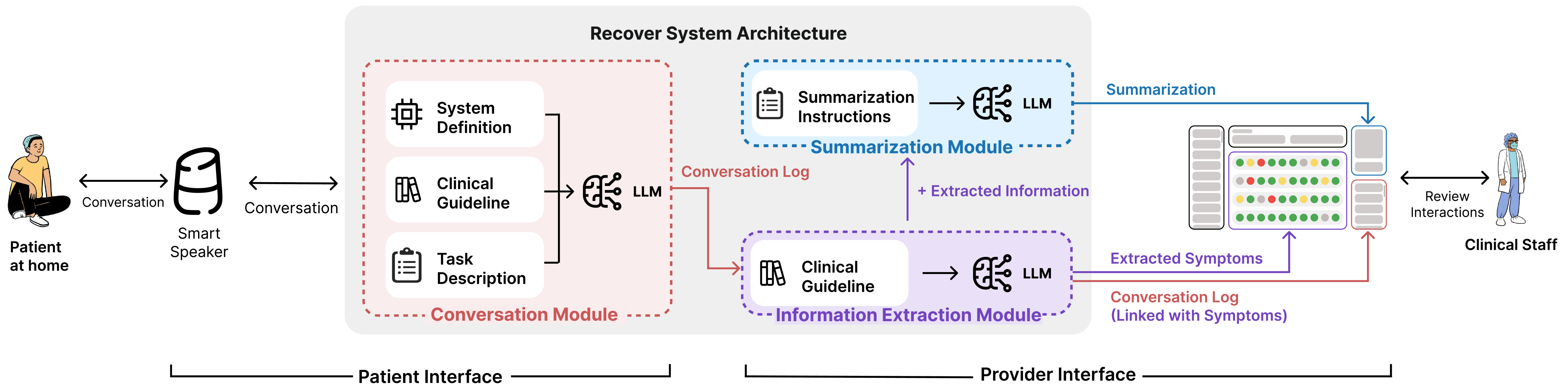
Session 7

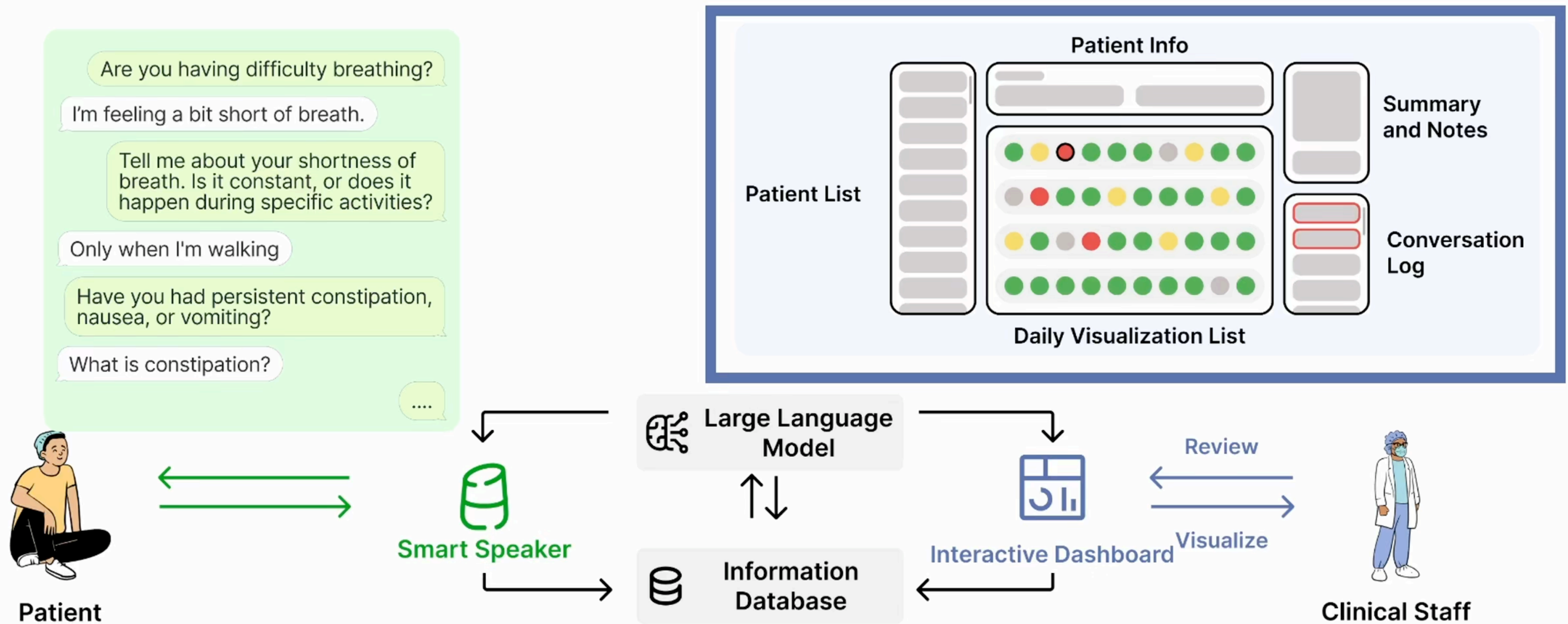
7 Design Sessions with Cancer Care Providers

RECOVER

What we did

- System Design and Development following the design strategies





RECOVER: Designing a Large Language Model-based Remote Patient Monitoring System for Postoperative Gastrointestinal Cancer Care

RECOVER

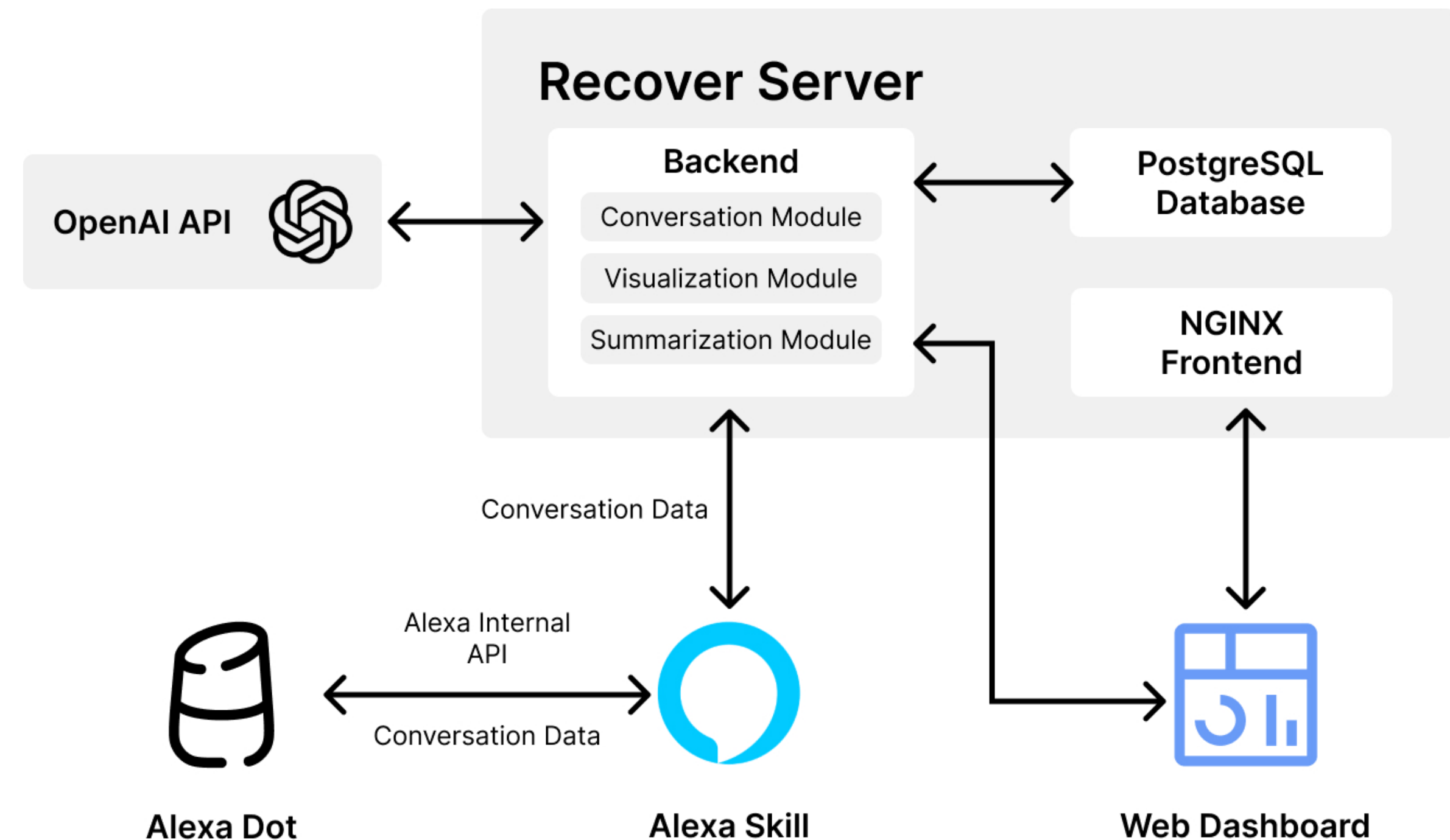
What we did: Evaluation

- Qualitative:
 - Invited cancer care providers to review the dashboard and a demo of VA interaction
 - Cancer patients to test the VA use and reflect on their insights
- Quantitative
 - Deployment with 15 patients, collected conversation logs, rated and compared accuracy and topic convergence

RECOVER

Specifications

- Allow parallel users, about 15-20 patient users; 1 conversation per day
- PostgreSQL database, NGINX frontend
- To be hosted in hospital-managed web servers that are HIPPA-compliant
- 1 experienced developed over a few months



RECOVER

Why build it?

- Cancer healthcare providers wanted a way to easily monitor their patient conditions post-operatively
 - Design of visualizations with LLM
 - Design of highlights/matches with LLM
- We as researchers, would like to explore how to design LLM-powered remote patient monitoring systems that comply with clinical guidelines

RECOVER

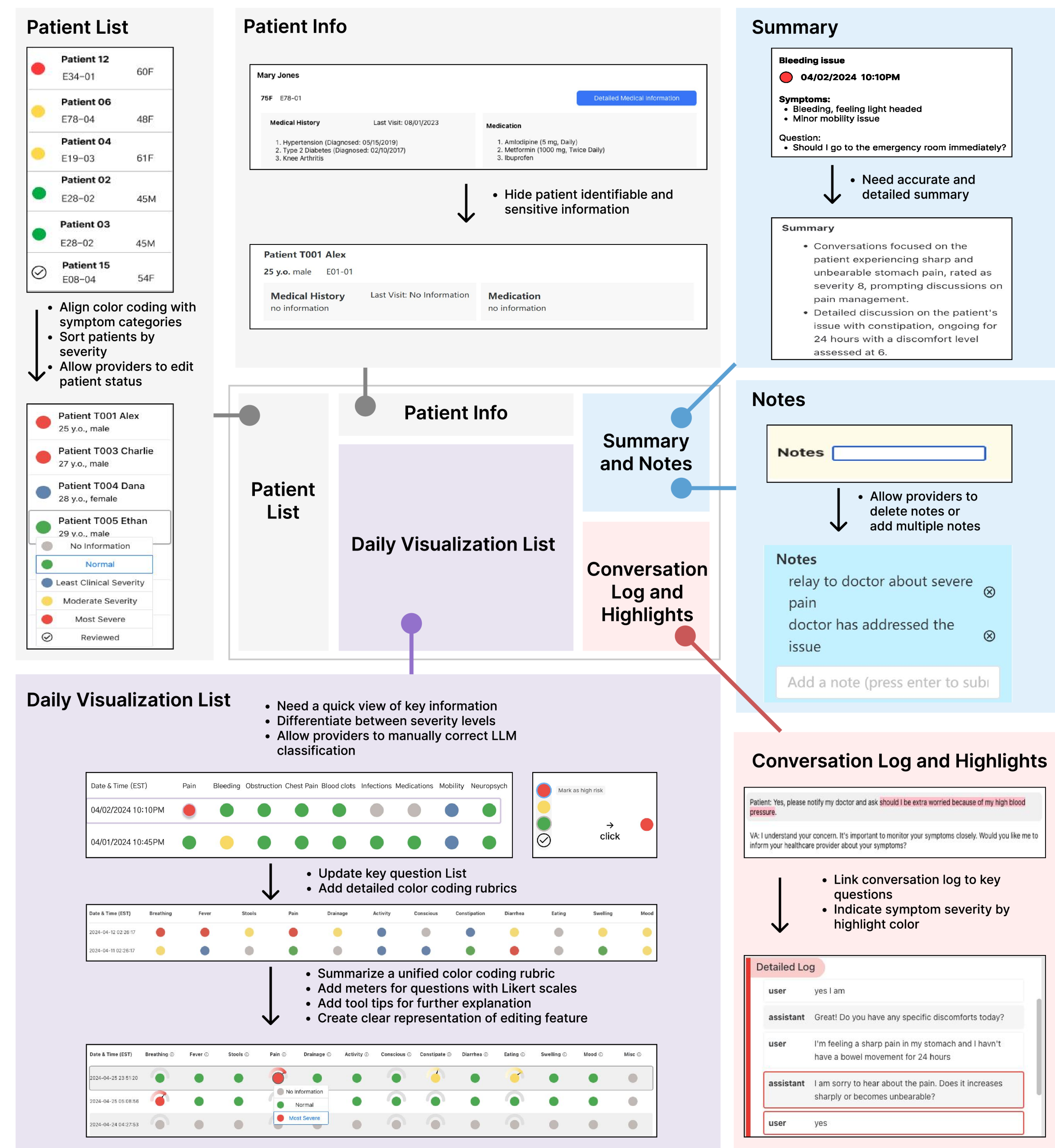
What was easy?

- Designing a first-draft for the prototype
- Developing the website, connect to a database
- Plan a pilot test for the system, based on previous experiences

RECOVER

What was hard?

- Figuring out what design is needed exactly, according to clinical practices
 - What color levels? Under what conditions?
 - What user operations/actions?
 - In what formats?
- Do prompt engineering for each module, iteratively with healthcare professionals



Question	Likert Scale	Sev
Are you having difficulty breathing?	Yes	Most
Are you having a fever of over 100 degrees, or chills?	No	Most
Have you had black, tar-like stools?	No	Most
Do you have pain that sharply increases, or becomes unbearable?	Yes	Most
Are you having any wound drainage problems,		

Daily Visualization List

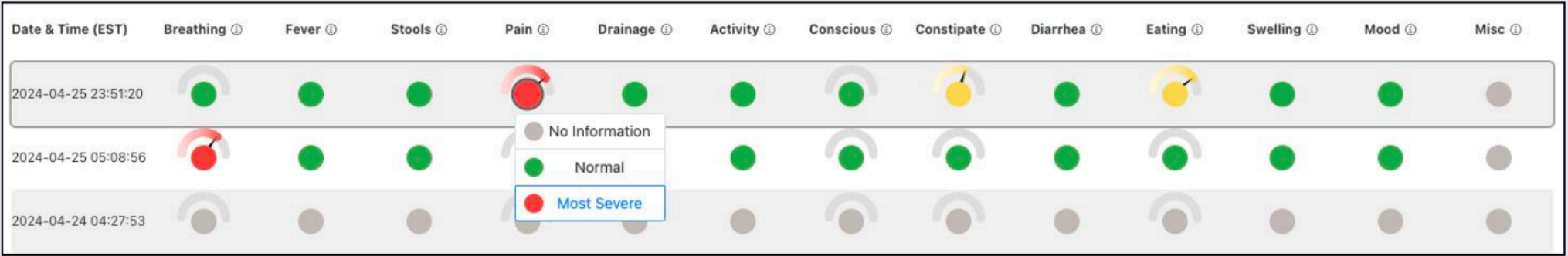
- Need a quick view of key information
- Differentiate between severity levels
- Allow providers to manually correct LLM classification



- Update key question List
- Add detailed color coding rubrics



- Summarize a unified color coding rubric
- Add meters for questions with Likert scales
- Add tool tips for further explanation
- Create clear representation of editing feature



A.7.2 Prompt for Information Visualization & Extraction Module.

Symptom Keys and Descriptions:

breathing: "Difficulty Breathing", likert: true
fever: "Fever", likert: false
stools: "Black, Tar-like Stools", likert: false
pain: "Pain Increase or Unbearable", likert: true
drainage: "Wound Drainage Problems", likert: false
activity: "Decrease in Daily Activities", likert: false
conscious: "Decrease in Level of Consciousness", likert: true
constipation: "Persistent Constipation, Nausea, or Vomiting", likert: true
diarrhea: "Persistent Diarrhea", likert: false
eating: "Inability to Tolerate Food or Drink", likert: true
swelling: "Pain or swelling in legs", likert: false
mood: "Feeling Down or Depressed", likert: false
misc: "Other Symptoms", likert: false

For each symptom listed above, perform the following tasks:

Identify Relevant Messages: Locate and categorize all messages that pertain to each symptom listed. This includes the assistant asking questions (e.g., "Are you having difficulty breathing?") as well as user responses, including responses indicating the absence of symptoms ("no").
EACH SYMPTOM SHOULD HAVE AT LEAST TWO MESSAGE (QUESTION AND RESPOND).

Assess state: determine whether the patient reported to this symptom. if the patient didn't talked about the symptom, echo 0. If the patient mentioned that they have the symptom, write 2. otherwise write 1.
Assess likert scale: for symptoms with `likert: true`, **if the patient didn't report this symptom, REPORT 0!!!** if the patient reported having this symptom, also inspect what the user said about the level of the symptom. should be a interger between 1 and 10. if the symptom have `likert: false`, don't output this value.

Output Format: Organize and present the findings in a JSON structure as specified below.
Ensure the symptom keys in the output JSON match exactly with those from the provided list:
-

RECOVER

Major Takeaways

- Sometimes the “client”/main user does not know what they need :)
- Doing field studies helps to understand the complexity and little specific details that matters

Privacy Sandbox

Privacy Sandbox

Context

The "privacy paradox": users say **they care about privacy** but don't act on it

- System opaqueness makes it hard to see consequences of privacy choices

- Users lack experience with how their data actually affects what they see online

Intention - Behavior Gap

Can an empathy-based sandbox help users understand their privacy decisions? Design considerations and strategies to actual design decisions

Privacy Sandbox

Context

Can an empathy-based sandbox help users understand their own privacy decisions?

Privacy Sandbox

Our process

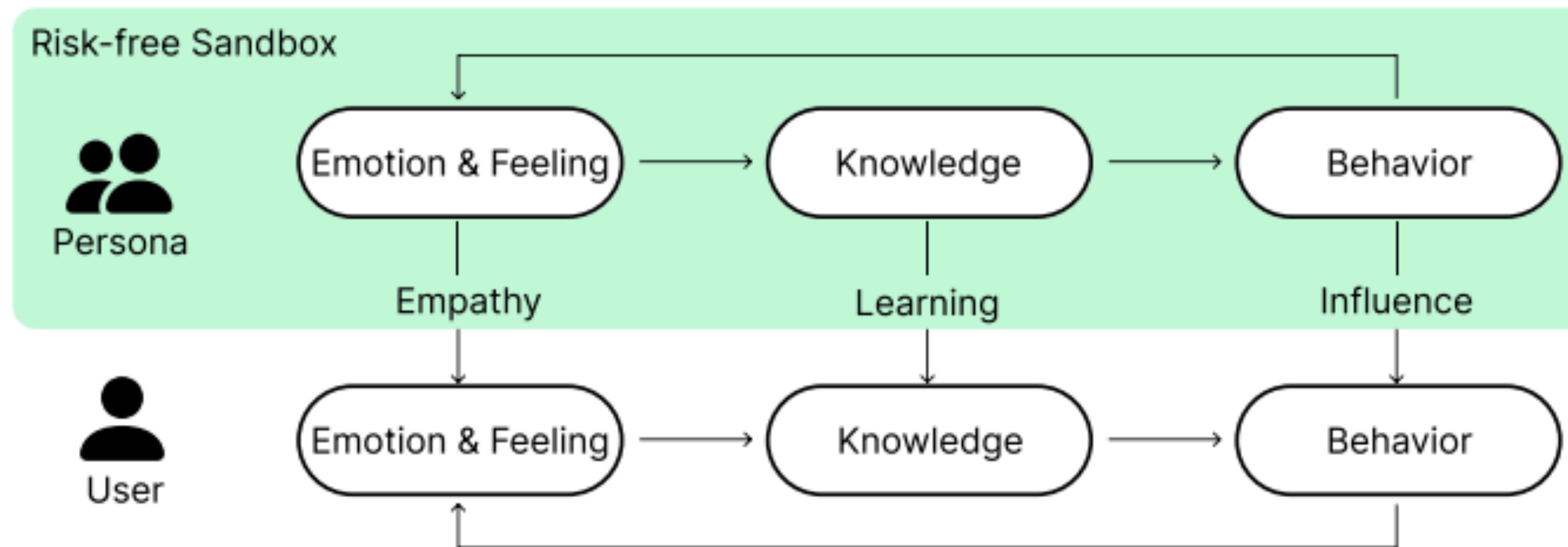
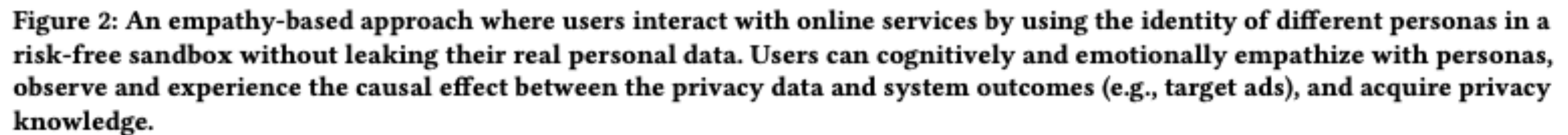


Figure 1: An empathy-based approach where users interact with online services with different personas in a risk-free sandbox without leaking their real personal data. Users can observe and experience the causal effect between their privacy configurations/behaviors and system outcomes, acquire privacy knowledge, and translate the knowledge into actual behavior. Note that this is a general framework. We leave the studies of changes in privacy behaviors as future work.

Our process



Privacy Sandbox

How it works

Users browse the web "as" an AI-generated **persona**

The sandbox replaces:

- Google account profile in Ad Center

- IP address (via NordVPN)

- Geographic location (OpenStreetMap)

- Browsing history (Chrome SQLite DB)

- User agent string

Users see how different personas get different targeted ads

LLMs (GPT-4) generate realistic persona data: **demographics, browsing history, social media**

Privacy Sandbox

What was hard?

Getting all the pieces to work together

VPN + geolocation override + user agent spoofing + browsing history injection all had to be coordinated

Ensuring persona data felt "real" enough to trigger different ads

Required extensive prompt engineering and validation

Privacy Sandbox

What was easy?

- **Modular architecture made components reusable.**

Because each identity signal (IP, location, user agent, profile, history) was implemented as an independent module, we could plug in or swap components without touching the rest of the system.

- **LLMs made persona generation surprisingly straightforward**
GPT-4 + few-shot learning + chain-of-thought produced realistic data
- **Using existing VPN APIs simplified IP switching**

Privacy Sandbox

More detail

We implemented a coordinated identity-replacement pipeline that synchronizes multiple browser-level modifications before each session starts. **Specifically:**

- We automated Google Ad Center profile updates using Puppeteer and related plugins.
- We overrode browser geolocation using latitude and longitude derived from OpenStreetMap.
- We dynamically switched IP addresses via NordVPN's API based on the persona's current location.
- We modified the browser's user agent string using Puppeteer's setUserAgent.
- We injected synthetic browsing history by overwriting Chrome's local SQLite history database before launch.

All modifications are applied in a controlled initialization phase to ensure consistency across identity signals.

For persona realism, we used structured prompt templates to generate **demographic attributes and browsing patterns**, and validated personas by observing whether they produced distinct downstream ad personalization outcomes.

Privacy Sandbox

Generated persona data

 **Device and browser settings** ☐

Chrome - Mac

Generate Device

Mozilla/5.0 (Macintosh; Intel Mac OS X 10_14_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/114.0.0.0 Safari/537.36

 **Browser history** ☐

Select Time

2023-07-11

→

2023-07-17

Count

353

Generate Browser History

17:51:15	Latest Fashion Trends for Summer 2023 - Google Search	https://www.google.com/search?q=latest+fashion+trends+summe...	⋮
15:21:01	Fashion Trends for Summer 2023 - The Ultimate Guide	https://www.fashionmagazine.com/summer-2023-fashion-trends/	⋮
15:21:00	Business Administration Degree - Bachelor's Program - University...	https://www.utsa.edu/business/academics/undergraduate/majors/...	⋮
13:33:49	Best Project Management Tools - Google Search	https://www.google.com/search?q=best+project+management+to...	⋮
13:33:49	Fashion Trends for Summer 2023 - Shop the Latest Styles - Nord...	https://www.nordstrom.com/browse/trends/summer-fashion?origi...	⋮
13:28:32	Office Outfits - Google Search	https://www.google.com/search?q=office+outfits	⋮
13:23:15	Best wireless mouse for laptop - Google Search	https://www.google.com/search?q=best+wireless+mouse+for+lap...	⋮
13:23:15	Business Attire for Women - Google Search	https://www.google.com/search?q=business+attire+for+women	⋮
13:07:28	Latest fashion trends for women - Google Search	https://www.google.com/search?q=latest+fashion+trends+for+wo...	⋮
13:06:36	Top project management tools for business	https://www.businessnewsdaily.com/16657-best-project-manage...	⋮

 **Twitter posts** ☐

Select Time

2023-07-13

→

2023-07-19

Count

21

Generate Twitter Posts



2023-07-19 21:10:38

Getting lost in a captivating book tonight. Escaping to different worlds through the power of words! 📖 #Bookworm

📍 Reading Books - 8769 Birch Avenue, Seattle, WA 98106



2023-07-18 18:45:17

Savoring a scrumptious dinner with my loved ones. Nothing beats quality time and good food! 🍽️ #FamilyTime


📍 Dinner - 8765 Cedar Lane, Seattle, WA 98104

Add

 **Facebook posts** ☐

Select Time

2023-07-13

→

2023-07-19

Count

21

Generate Facebook Posts

Enjoying a delicious and nutritious lunch break. Fueling up for an afternoon of productivity! #HealthyEating


📍 Lunch - 2345 Walnut Street, Seattle, WA 98107

2023-07-17 08:30:12

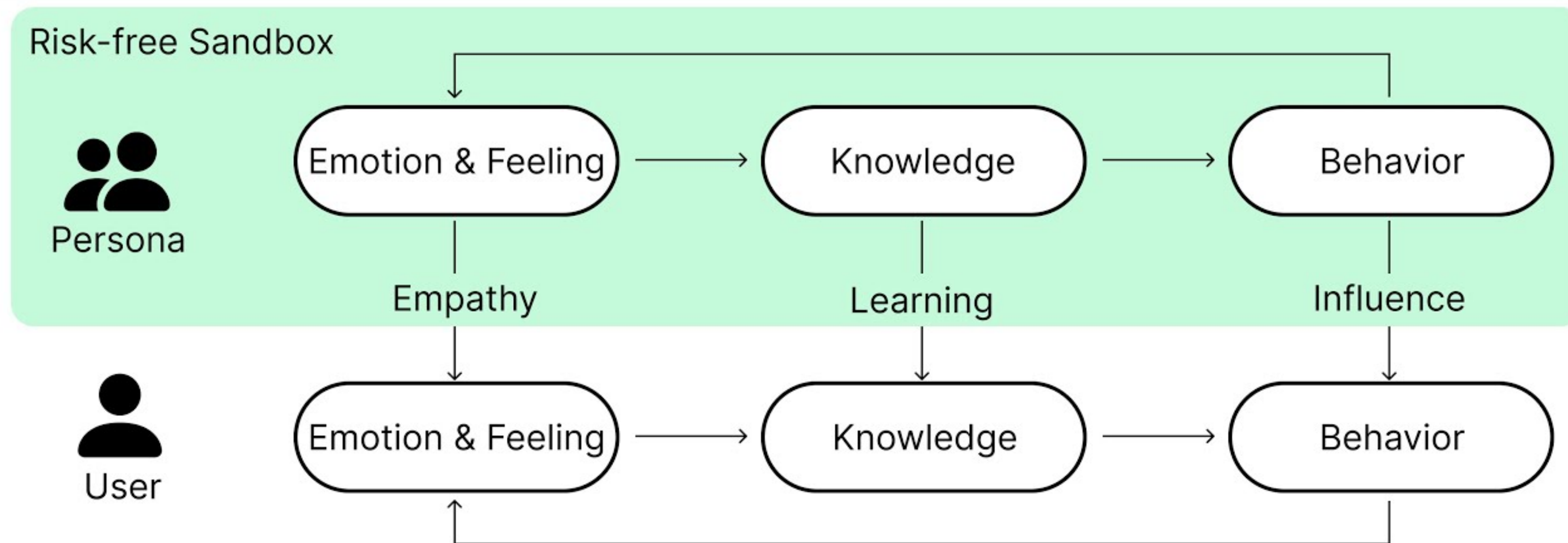
Started the day with a refreshing exercise routine. Feeling energized and ready to conquer the day! 💪 #MorningWorkout

📍 Morning Exercise - 5678 Maple Avenue, Seattle, WA 98102

Add

57

Privacy Sandbox



Privacy Sandbox

What does this teach us about prototyping?

Synthetic data is a prototyping superpower

LLM-generated personas replaced months of recruitment.

But you must validate fidelity.

Building on top of platforms is powerful but fragile

You get real-world behavior (actual ads!) but you're at the mercy of UI changes

The study worked: users changed their privacy settings after the experience

A stitched-together prototype produced real behavioral change

What have you learned from class today?

Nobody has responded yet.

Hang tight! Responses are coming in.

Overall Reflections

About Prototyping and Research

- Building a nice-looking prototype is usually not the only goal, but in the process...
 - You learn to adjust the prototype to real-world needs, or
 - You assess the feasibility of this design idea
- A prototype answers a research question, it doesn't necessarily solve the problem.
- Don't start from scratch, stand on the shoulders of existing systems.
- Deployment reveals what you didn't know you didn't know

Today's goals

By the end of today, you should be able to...

- Articulate requirements needed for developing early-stage prototypes
- Consider how development requirements often shape design needs, for better or worse
- Describe research outcomes related to personal informatics

IN4MATX 133: User Interface Software

Lecture 19: Prototyping in Research